Regime-Bound Measurement in Complex Systems: Proxy, Placement, and Validity

Morten Magnusson

Symbiose Research, Sandnes, Norway

ORCID: 0009-0002-4860-5095

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Abstract

Observed outputs in complex systems are not direct reads of the underlying phenomenon, but regime-conditioned products of proxy choice, instrument logic, observer placement, and interpretive compression. This paper formalises a general measurement framework in which analysis is *cartography*—mapping outputs as coordinates on a structured response surface—rather than verdict. We define five operative concepts: *regime* (the local rule-set governing valid inference), *proxy* (the mediating variable through which a phenomenon becomes measurable), *placement* (the observer's position in measurement and interpretive space), *episenter* (the weighted set of privileged variables organising a given analysis), and *compression* (the structural reduction inherent in any measurement act).

Using two empirical cases—cosmological structure analysis via WP3/$R(k, S)$ and regime-dependent validity in AI evaluation—we demonstrate that apparent conflicts between models are often regime mismatches rather than direct empirical contradictions. Proxy variables are not merely technical approximations, but constitute the primary mechanism by which systems gain and lose proximity to the phenomena they attempt to read. The framework does not deny the possibility of robust truth-claims; it specifies the conditions under which such claims become *regime-stable* rather than merely locally coherent. Testable predictions, falsification criteria, and a complete prediction register across both domains are derived from this framework.

Contents

1 The Problem: Measurement Read as Direct Readout

Most empirical frameworks treat observed output as an approximately direct representation of the underlying phenomenon. A model is specified, predictions are computed, residuals are assessed, and the measurement pipeline is assumed to be transparent. This assumption is rarely stated explicitly—which is precisely what makes it dangerous.

The assumption is not neutral. It imports three hidden commitments: that the measurement instrument has no structural effect on what is seen; that the observer's position in the measurement space does not change what is accessible; and that the same model can be validly evaluated across all conditions simultaneously. Each of these commitments fails in specific, diagnosable ways.

1.1 Three Named Failures

- **Regime conflation:** Applying a model across multiple conditions simultaneously without first establishing that the same inferential rules hold in all of them. If the underlying system exhibits regime structure—if response functions change with structural state—pooled analysis will systematically misread residuals at regime boundaries. The failure is not random: residuals at boundaries will be systematically larger and less coherent than residuals in regime interiors.
- **Proxy opacity:** Using a measurement variable as if it were a direct read of the underlying quantity it represents. Every measurement variable is a proxy— it makes a phenomenon accessible at the cost of structural blindness to the aspects it does not capture. Treating the proxy as the phenomenon is not an approximation error; it is a category error. The blindness is structural, not accidental.
- **Episenter universalism:** Treating one particular metric, parameter, or reference variable as the universal organising centre of the analysis. The choice of episenter determines what appears as primary signal and what appears as peripheral residual. An undeclared episenter is an invisible assumption with testable consequences: change the episenter, and the apparent structure of the data changes systematically.

1.2 The Positive Claim

This paper is not an argument that measurement is impossible or that truth is inaccessible. It is an argument for precision: measurement is mediated, and the mediation is structured and tractable. Making the structure explicit does not undermine analysis—it makes analysis more reliable, because it replaces implicit assumptions with declared choices.

Core epistemological claim: The framework does not eliminate measurement. It specifies the conditions under which measurement produces *regime-stable* truth-claims rather than locally coherent artifacts. A regime-stable truth-claim holds across multiple proxy configurations, placement variations, and episenter choices within a defined regime. This is a stronger, not weaker, standard of evidence than single-measurement confirmation.

1.3 Cartography as the Alternative

The alternative to direct-readout analysis is cartographic analysis: treating the measurement output as a coordinate on a response surface rather than as a verdict about the underlying phenomenon. This principle is developed in companion papers under the heading of the Regime-Consistent Measurement Principle [9] and Regime-Based World Modeling [13]; the present paper formalises the common measurement-theoretic structure across both.

A cartographic output reports: at coordinate (k, S) in the structured parameter space, under proxy $\hat{S}$, from instrument placement Π, the response is R. This output is simultaneously more honest—it declares its conditions—and more useful, because it can be compared to other cartographic outputs at the same coordinate under different conditions.

Null results become cartographic information: a null at coordinate (k_0, S_0) constrains the response function there. It does not generalise to other coordinates automatically. A null in the LATENT regime is not the same observation as a null in the FLOW regime, even if the scalar residual is identical.

2 Operative Definitions: Five Core Concepts

This section defines the five core concepts of the framework. Each definition is *operative*: it specifies what the concept picks out, how it differs from adjacent concepts, what its structural properties are, and what it would mean for the concept to fail.

2.1 Regime

Definition

A **regime** is a bounded domain in parameter space where a consistent set of inferential rules, response functions, and validity criteria holds. The regime-first epistemological discipline underlying this definition is formalised in the EBE Core Principles [8].

Three structural properties

1. *Interior stability:* Within the domain, the inferential rules hold uniformly. The same model, proxy, and episenter produce consistent results anywhere in the interior.
2. *Boundary conditions:* At the domain boundary, behaviour changes structurally. The transition may be sharp (a phase boundary) or gradual (a crossover region). In either case, measurements near the boundary have higher interpretive uncertainty than measurements in the interior.
3. *Non-universality:* Validity within one regime does not extend automatically to another. A model calibrated in one regime requires explicit re-validation before being applied in a different regime.

What a regime is not

A regime is not the same as a scale, a parameter range, or a redshift bin. Those are geometric subdivisions of the measurement space. A regime is defined by the local *physics* or *inference structure*—the rules that govern what can be validly concluded from measurements in that domain. Two adjacent parameter bins may be in the same regime; two non-adjacent bins may be in different regimes.

Cosmology: EFC defines FLOW (active energy redistribution, high ρ_{SFR}), TRANSITION (near-boundary dynamics, moderate ρ_{SFR}), and LATENT (suppressed response, low ρ_{SFR}) as structural regimes over $\hat{S}(z)$. These are not redshift bins—they are structural states of the cosmic matter distribution. A model valid in FLOW is not guaranteed valid in LATENT.

AI evaluation: A model validated on formal academic benchmarks occupies a different evaluation regime than the same model on conversational or adversarial inputs. High benchmark performance does not transfer automatically. The two regimes differ not just in difficulty but in the inferential rules that govern what "good performance" means.

2.2 Proxy

Definition

A **proxy** is a mediating variable through which a phenomenon is made measurable. Proxy choice simultaneously creates visibility into certain structural dimensions and structural blindness to others. This asymmetry is not incidental—it is the defining property of proxy use.

The visibility–blindness asymmetry

The asymmetry is structural: the proxy captures specific projections of the underlying phenomenon and necessarily discards others. The dimensions it discards are not random noise—they are structurally determined by the proxy construction. This means that the blindness can be characterised in advance: given a proxy's construction, one can specify which dimensions of the phenomenon will be inaccessible to it.

Regime-conditioned proxy validity

A proxy is valid within a regime; outside that regime, the same proxy may introduce systematic bias. The proxy's validity scope must be stated as part of the proxy declaration, not assumed to be universal.

Declared limitations requirement

The framework requires that every proxy be accompanied by an explicit statement of its declared limitations: the structural dimensions it cannot capture, the regimes in which it is known to be unreliable, and the conditions under which it is expected to break down.

Cosmology: $\hat{S}(z) = \log_{10} \rho_{\mathrm{SFR}}(z) + \log_{10} f_{\mathrm{gas}}(z) + C_0$ is a monotonic proxy for structural maturity S. *Visibility:* cosmic average structural state at redshift z. *Declared blindness:* local environment variations—a galaxy cluster and a void at the same z receive the same $\hat{S}$ value. Physical interpretability is limited by this construction.

AI evaluation: BLEU score is a proxy for translation quality. *Visibility:* n-gram overlap with reference translation within a defined evaluation regime. *Declared blindness:* semantic adequacy, pragmatic coherence, multi-reference validity, register variation, and cross-regime transfer performance.

2.3 Placement

Definition

Placement is the observer's operative position across three analytically independent spaces.

Three placement dimensions

(i) *Physical placement:* coordinate location in the physical or parameter space of the system being studied.

(ii) *Instrument placement:* the measurement device, protocol, or procedure that defines which aspects of the system are accessed and how the raw signal is converted to a datum.

(iii) *Interpretive placement:* the theoretical framework, model, and set of background assumptions within which results are interpreted and compared to predictions.

These three dimensions are independent: two observers at identical physical placement may reach different conclusions if their instrument or interpretive placements differ. Before comparing two measurements, placement compatibility across all three dimensions must be verified.

Cosmology: CMB and SNIa measurements of H_0 occupy different physical placements ($z \approx 1100$ vs. $z < 0.1$), different instrument placements (anisotropy power spectrum vs. distance-redshift relation), and different interpretive placements (early-universe FLRW extrapolation vs. local distance ladder calibration). These are not simply two measurements of the same quantity at different precision—they are measurements from three distinct placement configurations that each require explicit characterisation before comparison.

AI evaluation: A human expert rater and an automated metric (e.g., BLEU) evaluating the same model output occupy different instrument placements (subjective holistic judgement vs. n-gram overlap) and different interpretive placements (pragmatic quality vs. surface-form similarity). Conflicts between them are often placement conflicts, not empirical errors.

2.4 Episenter

A trivial example first

In an ordinary least-squares regression, the residual sum of squares is the episenter—it is the quantity being minimised, and everything else (coefficient estimates, significance tests, model comparisons) is organised around it. Changing the episenter from sum-of-squares to absolute deviations produces a different model with different sensitivity properties. Neither is wrong; they are different epistemic choices. The episenter concept generalises this observation: every analysis has an implicit or explicit centre, and the choice of centre determines what counts as signal and what counts as noise.

Definition

An **episenter** is the operative centre of a measurement system: the weighted set of privileged variables and interpretive choices that determines what appears central and what is pushed to the periphery.

Key property: constructed, not found

The episenter is not a natural feature of the data. It is constructed by the choices made in setting up the analysis: which variable is placed on the primary axis, which residuals are reported, which comparisons are made. These choices have testable consequences: two analyses of the same data with different episenter placements will produce structurally different output distributions.

Consequences of episenter choice

- What appears as primary signal: deviations from the expected value of the episenter variable.
- What appears as secondary or peripheral: residuals on non-episenter variables.
- What is excluded entirely: any dimension of variation that does not project well onto the episenter variable.

Cosmology (dry example): Analysis A uses redshift z as the primary axis (z-episenter). Analysis B uses structural maturity S as the primary axis (S-episenter). Same data, same physics, different residual structures—because the episenter differs. EFC anomalies that appear random in z-space are sign-coherent in S-space. This is not a different model; it is the same data viewed from a different episenter.

AI evaluation: Evaluation system A uses accuracy as the episenter. System B uses calibration error. A model with accuracy 0.87 and poor calibration appears highly competent under A and unreliable under B. Neither verdict is wrong— they reflect different episenter placements applied to the same model.

2.5 Compression

Definition

Compression is the structural reduction inherent in any measurement act: the mapping from a high-dimensional phenomenon to a lower-dimensional output.

Key properties

- *Necessary:* Compression is not an error; it is a necessary condition for measurement. Without reduction, there is no signal—only raw, unstructured data.
- *Selective:* Compression preserves some structural properties and discards others. The choice of what is preserved is determined by the proxy construction and the episenter placement.
- *Consequential:* The loss is not random noise. It is structured: the dimensions discarded by compression are structurally determined by the measurement design, and their absence has predictable consequences for what can be concluded.

The analytical question

The question for any measurement system is not whether compression occurs—it always does—but which structural properties are retained and which are lost, and how consequential the loss is for the intended inference.

Cosmology: $P(k)$ compresses the full 3D density field into a 1D function of wavenumber. It retains second-order statistics (two-point correlations) but discards phase information and higher-order correlations (the Fourier phases encode the non-Gaussian structure of collapsed objects). Two density fields with identical $P(k)$ can have radically different visual and structural properties.

AI evaluation: A scalar benchmark score (e.g., 0.84 on MMLU) compresses model behaviour across an entire evaluation distribution into one number. It retains mean correctness but discards output variance, tail behaviour, failure mode distribution, calibration quality, and cross-regime consistency. Two models with identical scores may have radically different deployment-relevant behaviour profiles.

2.6 Reference Table

Concept	Operative definition	Failure mode	Test
Regime	Bounded domain where consistent inferential rules hold	Conflation: pooling across regime boundaries	Residuals by regime differ significantly
Proxy	Mediating variable with defined visibility and declared blindness	Opacity: treating proxy as direct read	Proxy substitution produces systematic drift
Placement	Position across physical, instrument, interpretive space	Incompatibility: comparing across placement dimensions	Placement-matching reduces apparent conflict
Episenter	Weighted privileged variables organising what appears central	Universalism: undeclared single episenter	Episenter shift produces structured redistribution
Compression	Structural reduction from phenomenon to output	Neglect: scalar treated as complete representation	Lost dimensions have predictable footprint

Table 1: Five operative concepts with definitions, failure modes, and associated testability criteria. Each failure mode has a corresponding prediction (Section 8).

2.7 Structural Dependencies

The five concepts are not independent. Their structural relationships are:

1. **Regime grounds validity:** All other concepts are regime-scoped. A proxy, placement, episenter, and compression account are valid only within a declared regime. Regime identification is the prerequisite for all other framework operations.
2. **Proxy implements placement:** Instrument placement determines which proxies are accessible. Changing the instrument placement changes the available proxy set. A proxy choice and an instrument placement are therefore not independently chosen.
3. **Episenter organises compression:** What compression retains is shaped by the episenter. An accuracy-episenter system will retain accuracy-relevant information and discard calibration-relevant information. The compression loss is not random—it is structured by the episenter choice.
4. **Compression produces output:** The final observed datum is the result of all preceding choices: regime-scoped, proxy-mediated, placement-conditioned, episenter-organised, and compression-filtered. The output is not a direct read of the phenomenon. It is the product of a structured chain of mediation.

The core structural chain:

$$\underbrace{\text{regime}}_{\text{scope}} \longrightarrow \underbrace{\text{proxy} + \text{placement}}_{\text{access}} \longrightarrow \underbrace{\text{episenter}}_{\text{organisation}} \longrightarrow \underbrace{\text{compression}}_{\text{reduction}} \longrightarrow \underbrace{\text{output}}_{\text{observed datum}}$$

Each arrow is a site of epistemic choice. Making these choices explicit is the methodological core of the framework.

3 Formal Thesis and Derived Predictions

Core thesis: Observed outputs in complex systems are not direct reads of the underlying phenomenon, but regime-conditioned products of proxy choice, instrument logic, observer placement, and interpretive compression.

This thesis has four components. First, it denies that observed outputs are direct. Second, it specifies what they are instead: regime-conditioned products. Third, it identifies the four mediating factors: proxy, instrument, placement, and compression. Fourth, it implies testability: if the mediating factors structure the output, then manipulating them should produce predictable and systematic changes.

The thesis does not deny that truth-claims are possible. It specifies what a *regime-stable* truth-claim is: one that holds across multiple proxy configurations, placement variations, and episenter choices within a defined regime. This is a stronger, not weaker, standard than single-measurement confirmation.

3.1 Nine Theses

1. **Measurement is not direct readout.** Direct readout requires explicit justification, not default assumption. The burden of proof is on the claim that the observed output reflects the underlying phenomenon without structural mediation.

2. **Proxy creates both visibility and blindness.** Proxy choice is an epistemic choice. The structural blindness of a proxy is as important as its visibility, and must be declared alongside the proxy construction.

3. **Measurement placement is a primary question.** Three distinct spaces (physical, instrument, interpretive) must be characterised before any comparison between measurements can be meaningful.

4. **Episenter is produced, not given.** The episenter is a weighted set of privileged variables, constructed by analysis choices, not found in the data. Different episenter placements produce different apparent structures in the same data.

5. **Conflicts may be regime conflicts.** Regime identification is a prerequisite for meaningful comparison. An apparent empirical conflict that dissolves when the measurement regimes are identified is a placement conflict, not a physics conflict.

6. **Validity is regime-conditioned.** Universal validity claims require regime-robustness assessment. A model valid in one regime requires explicit re-validation before deployment in a different regime.

7. **Analysis of the analysis is part of the analysis.** Omitting scrutiny of instrument, proxy, and placement is not neutral—it is an undeclared epistemic choice with predictable consequences. The analysis of the measurement system is not a meta-level activity; it is internal to the analysis.

8. **Measurement is compression.** The question is not whether structural reduction occurs—it always does—but which structural properties are preserved vs. lost, and how consequential the loss is.

9. **A fragment carries holistic structure under four conditions:** (a) structural logic preservation (the fragment follows the same inferential rules as the whole), (b) episenter compatibility (the fragment's episenter is compatible with the whole's), (c) directional

consistency (the fragment's deviations are directionally coherent with the whole's), and (d) regime compatibility (the fragment and the whole are in the same regime).

3.2 Five Derived Predictions

1. **Consistent response at shared coordinates (P1):** Independent probes mapped to the same coordinate in (k, S) space should produce consistent responses. Inconsistency at matched coordinates is diagnostic of regime mismatch, instrument incompatibility, or proxy divergence—not random noise.

2. **Systematic proxy-substitution drift (P2):** Replacing one proxy with a structurally different proxy produces directional, predictable drift in output—not random scatter. The direction is determined by the proxy's declared structural blindness. This is testable: the drift direction can be predicted before the substitution is run.

3. **Regime-resolution of conflicts (P3):** Conflicts resolved by regime identification but not by model substitution provide positive evidence for the framework. If recognising that two measurements come from different regimes dissolves the apparent contradiction, that is structure, not coincidence.

4. **Cartographic null results (P4):** Null results carry regime-specific information. A null in the LATENT regime constrains the response function there; it does not generalise to FLOW or TRANSITION. Null results are coordinates, not absences.

5. **Episenter-shift redistribution (P5):** Changing the episenter from one metric to a structurally different one produces systematic, predictable redistribution of apparent rankings or residuals—not random reordering. The redistribution direction is predictable from the structural relationship between the two episenter choices.

4 Case 1: Cosmology (WP3 / $R(k, S)$)

4.1 Standard Reading and Its Epistemic Cost

Standard cosmological analysis asks: does the model fit the data? It fits a FLRW background, computes predicted observables, computes residuals, and assesses goodness-of-fit. The measurement pipeline is assumed transparent: the observed quantity is treated as a direct read of the underlying cosmological parameter.

Three specific epistemic failures follow from this assumption in the current cosmological context:

- **Regime conflation:** Fitting a single FLRW background across all redshifts simultaneously without testing whether the same inferential rules hold throughout. If the universe exhibits structural regime transitions—as EFC predicts—pooled fitting will systematically misread residuals at regime boundaries as model-level tensions.
- **Proxy opacity:** Using z as a direct proxy for cosmic time and structural state without declaring its limitations. Redshift is a kinematic proxy. It encodes expansion history, not structural state. Two redshifts with identical z but different structural configurations (e.g., one in a dense filament and one in a void) will be treated identically by a z-based analysis.
- **Episenter universalism:** Placing H_0 at the episenter of all cosmological inference. H_0 is a compression of the full expansion history into a single number. Two models with identical H_0 can have systematically different predictions for structure growth, lensing power, and baryon acoustic oscillation scaling.

4.2 The $R(k, S)$ Response Surface

The WP3 analysis replaces the verdict framework with a cartographic framework [4, 5]. Instead of asking "does the model fit?", it asks: "what is the local response structure at coordinate (k, S) in the parameter space, and is that response consistent with the predicted EFC regime structure?"

The response surface $R(k, S)$ maps:

- *Scale coordinate k:* the wavenumber of the structure probe (BAO scale $k \approx 0.03$–$0.15\,h\,\mathrm{Mpc}^{-1}$).
- *Structural state coordinate S:* the EFC regime, indexed by $\hat{S}(z)$.

A measurement at (k_0, S_0) is a coordinate on this surface. Its deviation from the FLRW prediction at that coordinate is a cartographic datum. A null deviation is not "FLRW works here"—it is a data point at $(k_0, S_0, \Delta R = 0)$. A significant deviation is a data point at $(k_0, S_0, \Delta R \neq 0)$.

4.3 The Three EFC Structural Regimes

Structural maturity S partitions observable redshift space into three regimes over $\hat{S}(z) = \log_{10} \rho_{\mathrm{SFR}}(z) + \log_{10} f_{\mathrm{gas}}(z) + C_0$:

4.4 Proxy $\hat{S}(z)$: Construction and Declared Limitations

Construction

$$\hat{S}(z) = \log_{10} \rho_{\mathrm{SFR}}(z) + \log_{10} f_{\mathrm{gas}}(z) + C_0 \tag{1}$$

where:

Regime	$\hat{S}$ tertile	Physical character	Expected response
FLOW	Bottom	High ρ_{SFR}; active energy redistribution; rapid structural assembly	High response amplitude, higher variance
TRANSITION	Middle	Near-boundary dynamics; moderate ρ_{SFR}; regime crossover	Intermediate; high boundary sensitivity
LATENT	Top	Low ρ_{SFR}; suppressed response; structural equilibration	Near-null signal; coherent sign structure

Table 2: Three EFC structural regimes. Boundaries set by tertile quantiles over $z \in [0.07, 2.4]$ on a uniform grid, locked before analysis. Regime boundaries are not free parameters—they are derived from $\hat{S}(z)$ with no post-hoc adjustment.

- $\rho_{\mathrm{SFR}}(z)$ follows the Madau–Dickinson parametrisation.
- $f_{\mathrm{gas}}(z)$ uses Tacconi/PHIBSS Table 3b with $\beta = 2$.
- Equal weight for both components; C_0 is a normalisation constant.
- Locked parametrisation—no refit to the data.

Proxy scope

$\hat{S}(z)$ is a monotonic proxy for the *cosmic average* structural maturity at redshift z. It encodes the global star-formation and gas-fraction history of the universe.

Declared proxy limitation: $\hat{S}(z)$ encodes cosmic averages, not local structural state. A galaxy cluster and a void at the same z receive the same $\hat{S}$ value. The proxy cannot distinguish local environments, merger states, or halo mass distributions. Physical interpretability at sub-cosmic scales is limited by this construction. This limitation is methodologically owned and does not invalidate the regime test—it scopes it to the cosmic-average regime structure.

4.5 Axiom 0 Test: Design and Results

Test design

The Axiom 0 test evaluates whether the sign and magnitude of deviations from the FLRW prediction cluster near EFC regime boundaries in $\hat{S}$-space.

- **Primary statistic:** Mean distance of observed deviations from the nearest regime boundary, compared to a permutation null.
- **Secondary statistic:** Sign coherence of deviations within the TRANSITION + LATENT regime—fraction of deviations with consistent sign direction.
- **Dataset:** 10 BAO deviations from FLRW baseline, drawn from the literature (DESI DR2 and precursors).
- **Pre-registration:** Test design locked before analysis.

Independently, the EFC growth-sector signal has been tested under leave-one-out, sound-horizon, and amplitude-prior controls, producing a robust $\sim 2\sigma$ preference for $\alpha < 0$ in the Hubble-friction channel [7]. This is structurally consistent with the regime interpretation of the Axiom 0 secondary result.

Statistic	Result	Interpretation
Primary: mean distance to boundary	$p = 1.0$	Degenerate. Deterministic $\hat{S}$ mapping with $N = 10$ leaves no permutation variance. This is an informative methodological finding: the primary test as designed cannot discriminate EFC from the null at this sample size. Not a null result against EFC.
Secondary: sign coherence (TRANSITION + LATENT)	$p = 0.020$	8 of 9 points have positive $\hat{S}$-deviation direction. Consistent with EFC regime structure. Not pre-registered as primary; cannot be treated as confirmatory.
DESI $z = 1.0$ boundary distance	dist $= 0.000$	Sits exactly on the FLOW/TRANSITION boundary. Anecdotal at $N = 10$; recorded as a cartographic observation.
eBOSS LRG $z = 0.51$	2.0σ, dist $= 0.019$	Near TRANSITION/LATENT boundary. Consistent with boundary-sensitivity prediction.
$f\sigma_8$ $z = 0.7$	2.0σ	In TRANSITION regime. Consistent with high boundary-sensitivity prediction for TRANSITION.

Table 3: Axiom 0 test results (2026-02-20, $N = 10$ BAO deviations). The secondary statistic is consistent with EFC regime structure at $p = 0.020$ but requires $N \geq 30$ and pre-registration as primary statistic to be actionable. All results are preliminary.

What Axiom 0 establishes

The primary test failure is itself informative: with a deterministic proxy mapping and $N = 10$, the permutation distribution is degenerate. This is a methodological finding about the test design, not evidence against EFC. The secondary sign-coherence result ($p = 0.020$, $N = 10$) is consistent with EFC but is not confirmatory. Axiom 0 v2 (Section 8) pre-registers sign-coherence as the primary statistic with $N \geq 30$.

Placement dimension	CMB measurement	SNIa measurement
Physical (z)	$z \approx 1100$ (last scattering)	$z < 0.1$ (local universe)
$\hat{S}$-regime	FLOW (maximum ρ_{SFR})	LATENT (minimum ρ_{SFR})
Instrument	CMB anisotropy power spectrum	SNIa distance modulus ladder
Interpretive	FLRW extrapolation from $z = 1100$	Local calibration + FLRW
Compression	Full CMB power spectrum $\to H_0$	Full distance ladder $\to H_0$

Table 4: Placement analysis of the two primary H_0 measurement configurations. All three placement dimensions are incompatible. Under the framework, comparing these measurements without placement correction is methodologically premature.

4.6 The Hubble Tension as Regime Mismatch

The standard framing

The Hubble tension is the 4–5σ disagreement between H_0 measured from early-universe probes (CMB: $H_0 = 67.4 \pm 0.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, Planck 2018) and late-universe probes (distance ladder: $H_0 = 73.0 \pm 1.0\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, SH0ES). It is widely framed as requiring new physics.

The placement analysis

CMB and SNIa measurements of H_0 occupy radically different placement configurations (see also [6] for systematic CMB+LSS parameter-space localization):

The regime reframing

Under the EFC framework, H_0 is a regime-conditioned observable, not a universal constant with one true value. This reframing was first proposed in [10]. The CMB measures the FLOW-regime expansion rate; SNIa measures the LATENT-regime expansion rate. If EFC regime structure is real, these two measurements are cartographic coordinates at different regime locations, not two measurements of the same quantity.

The framework prediction (C4, Section 8): H_0 estimated within a single $\hat{S}$-regime should show lower inter-survey variance than H_0 estimated pooled across regimes.

4.7 Null Results as Cartographic Information

Three illustrative examples of the cartographic reading of null results:

1. **DESI DR2 background constraint:** $\alpha \approx 0.0 \pm 0.23$ from multi-probe MCMC (BAO + $f\sigma_8$ + H_z + SNIa, 69 data points, DESI DR2). Standard reading: EFC has no background effect, α is excluded at the $\sim 3\sigma$ level. Cartographic reading: EFC is not a background-level deformation—the null eliminates EFC as a global Friedmann-deformation while leaving the growth/regime-structure interpretation fully open. This null is informative: it *locates* EFC in the growth sector, not the background sector.
2. **Primary Axiom 0 $p = 1.0$:** Not "EFC predicts nothing". The test reveals a methodological limit of the primary statistic design. This is a cartographic coordinate in the space of test designs: at this sample size, the primary statistic has no discriminating power.
3. **KT2 $C = 2.32$ (convergent):** The EFC discrete-grid convergence parameter C

converges at $N = 81$ grid points. This null (no divergence) is a positive cartographic result: it establishes that EFC survives discretisation at this resolution level.

4.8 Honest Scope Statement

What is established by Case 1: (1) $\hat{S}(z)$ defines a testable, locked regime partition with no free parameters post-specification. (2) Axiom 0 secondary sign-coherence is consistent with EFC regime structure at $p = 0.020$ ($N = 10$, secondary statistic only). (3) DESI $z = 1.0$ sits exactly on the FLOW/TRANSITION boundary—a cartographic coincidence that requires larger N to interpret. (4) The Hubble tension is a placement-incompatible comparison under this framework.

Independently validated in the EFC programme (prior work, citable): The BAO transfer test (DESI → BOSS/eBOSS, no refit) is completed and passed at T2 [1]—establishing that the EFC growth/geometry sector transfers across survey boundaries without parameter re-tuning. This is a validated structural constraint, not exploratory. The Structural Coherence Evaluation (SCE) across KiDS, BAO, RSD, Lyα, and CMB lensing is completed at T4 [3], demonstrating higher layer consistency and lower model plasticity for EFC than for ΛCDM. The four-channel background coupling architecture test (BAO + RSD + SN Ia + CMB lensing, frozen β) is completed at T1 [2]. These results are relevant to the regime-bound measurement framework: they confirm that the EFC regime structure produces consistent, transfer-stable inferences— the operational definition of a regime-stable truth-claim.

What is not established by Case 1 of this paper: A detection-level signal from the Axiom 0 test. All results within this paper's worked example are exploratory and preliminary at current sample size. Regime identification of the Hubble tension is a reframing, not a resolution. The cartographic approach does not prove EFC—it specifies the conditions under which a proof-level claim would be possible. For the full validated EFC evidence base, see the companion validation ledger.

4.9 Testable Predictions from Case 1

C1 Same (k, S) coordinate → consistent response across independent probes (SPARC175/WP3). *Status: PARTIAL ($p = 0.020$, $N = 10$, secondary).*

C2 Proxy substitution from global $\hat{S}$ to a local structural proxy yields directional drift; direction predictable from declared blindness. *Status: PENDING (test designed; local proxy not yet available).*

C3 Largest BAO/growth deviations cluster near FLOW/TRANSITION and TRANSITION/LATENT boundaries in $\hat{S}$-space. Full DESI Year-5 test. *Status: ACTIVE.*

C4 H_0 estimated within one $\hat{S}$-regime has lower inter-survey variance than cross-regime pooled estimate. *Status: ACTIVE.*

C5 LATENT-regime residuals are sign-coherent; FLOW-regime residuals are higher-variance; LATENT/FLOW variance ratio > 2.0. *Status: PARTIAL.*

C6 Models with identical χ^2_{BAO} but different proxy constructions diverge in $f\sigma_8$ predictions at TRANSITION-regime coordinates. *Status: PENDING (Euclid WL).*

5 Case 2: AI Evaluation (Validity-Aware AI)

The validity-aware evaluation framework was introduced in [12]. The present paper provides its measurement-theoretic grounding and demonstrates the structural parallel with the cosmological case.

5.1 Standard AI Evaluation and Its Epistemic Cost

Standard AI model evaluation proceeds as follows: a model is trained, a benchmark dataset is selected, performance metrics are computed, and results are compared to baselines. High benchmark scores are treated as evidence of high capability. This procedure implicitly treats benchmark output as a direct read of the model's underlying capability.

Three specific epistemic failures follow:

- **Explainability conflation:** Explainability techniques (SHAP values, attention maps, saliency scores) report which input features most influenced the model's output in a given case. They do not report whether the model's reasoning is valid, whether its output is reliable across the evaluation regime, or whether its performance will generalise. Explainability is a measurement of the model's internal routing—it is not a measurement of the model's validity.
- **Benchmark regime opacity:** Benchmark datasets are drawn from specific distributional regimes—particular domains, registers, task structures, and annotation conventions. A model's performance on the benchmark is a regime-conditioned measurement. When the evaluation regime differs from the deployment regime, benchmark scores are not informative about deployment performance. The regime of the benchmark is rarely declared.
- **Episenter universalism:** Most benchmark systems place accuracy or F_1 at the episenter. Capabilities not well-projected onto this metric—calibration, coherence under distribution shift, failure mode structure, output variance—are classified as secondary or excluded. The episenter choice is rarely declared.

5.2 Explainability Is Not Validity

The distinction

Explainability reports which features of the input drove the model's output in a specific case. It answers: "what did the model attend to?"

Validity is the degree to which the model's outputs are accurate, reliable, and appropriately scoped within a defined evaluation regime. It answers: "does the model produce correct outputs, and under what conditions?"

These are measurements of different things. A model can be highly explainable and invalid. A model can be poorly explainable and valid within a well-defined regime. Treating high explainability as evidence of validity is a proxy conflation error: it substitutes a measurement of internal routing for a measurement of output quality.

Why the conflation persists

Explainability is operationalisable at scale: it can be computed automatically for any model on any input. Validity requires regime specification and scope declaration, which are more expensive to produce. The practical convenience of explainability is real. The epistemic cost—substituting an accessible proxy for the harder-to-measure target quantity—is equally real.

Concrete failure case: A medical image classifier that consistently attends to the correct anatomical region (high explainability: SHAP scores on correct features) may still produce miscalibrated probability scores for clinical decision-making (low validity: overconfident in low-prevalence conditions). The explainability measurement does not detect the calibration failure because calibration is not in its measurement scope.

5.3 Regime-Dependent Validity in AI Systems

Definition

An AI model's validity is regime-conditioned: it holds within a specific evaluation regime and should not be assumed to transfer across regime boundaries without explicit testing.

Four regime dimensions

Regime dimension	High-validity example	Low/unknown-validity example
Input distribution	Formal academic text matching training distribution	Informal conversational text, code-switching, low-resource languages
Task structure	Closed-form QA with single correct answer	Open-ended generation with multi-dimensional quality criteria
Annotation regime	Expert-labelled data in target domain	Crowdsourced labels, different protocol, different population
Deployment context	Batch processing in controlled environment	Real-time interactive deployment under adversarial input pressure

Table 5: Four dimensions defining an AI evaluation regime. A model validated in the left column has an *unknown* validity profile in the right column—not a low profile. Unknown is the correct epistemic state without explicit testing.

Regime boundary signals

Three diagnostic signals that a regime boundary has been crossed:

- **Performance variance increases:** Output variance typically increases near regime boundaries—structurally analogous to the higher interpretive uncertainty near FLOW/TRANSITION boundaries in the cosmological case.
- **Failure mode structure changes:** The character of errors changes at regime boundaries, not just their rate. Different error types signal a structural change in the evaluation conditions.
- **Metric stability breaks:** A metric reliable within one regime shows reduced correlation with human judgement when applied outside it.

Metric	Target quantity	Visibility scope	Structural blindness
BLEU/ROUGE	Translation / summarisation quality	n-gram overlap with reference; reliable in formal, single-reference regimes	Semantic adequacy, pragmatic coherence, multi-reference validity, register variation
Accuracy/F_1	Classification correctness	Mean correctness on i.i.d. held-out set	Calibration, tail behaviour, failure mode distribution, OOD performance
Perplexity	LM quality	Log-likelihood on test set; sensitive to distributional fit	Downstream task performance, factual accuracy, coherence over long contexts
Human preference ratings	Overall output quality	Captures pragmatic and contextual quality; regime-flexible	Rater variance, annotation protocol dependence, demographic coverage
SHAP / attention	Feature attribution (explainability)	Local attribution for single inference; mechanistically informative	Global validity, calibration, cross-regime reliability, output accuracy

Table 6: Proxy structure of standard AI evaluation metrics. Every metric has a defined visibility scope and a structural blindness cost. The blindness is structural, not random.

5.4 Proxy Structure of Evaluation Metrics

The compression cost of scalar metrics

Every scalar metric compresses the full distribution of model behaviour into a single number. Two models with identical scalar scores may have radically different behaviour profiles. The scalar metric cannot distinguish them. This is Thesis 8 applied to AI evaluation: the question is not whether compression occurs, but which structural properties are retained and which are lost.

5.5 Episenter and Placement in AI Evaluation

The episenter problem

Every AI evaluation system has an episenter. In most standard benchmarks, accuracy or task-specific F_1 occupies the episenter. Consequences:

- **Primary signal:** differences in accuracy across models.
- **Peripheral:** calibration quality, output variance, cross-regime transfer, failure mode structure, distributional fairness.
- **Excluded:** any capability dimension that does not project well onto the accuracy metric.

Three placement dimensions in AI evaluation

Input placement (physical analogue): Which input types, domains, and difficulty levels are included. A model evaluated only on easy, in-distribution inputs is in a different input placement than one evaluated on adversarial or OOD inputs.

Instrument placement: Which evaluation protocol is used—automated metric, human raters, red-teaming, deployment monitoring. Each accesses different aspects of model behaviour.

Interpretive placement: Which theoretical framework is used to interpret results—capability-first, safety-first, calibration-first, task-performance-first. Two evaluators with identical instrument placements but different interpretive placements will produce systematically different assessments of the same model.

5.6 Structural Parallel Across Both Domains

The structural parallel is the central argument for the framework's generality.

Concept	Cosmology (Case 1)	AI Evaluation (Case 2)
Regime	FLOW/TRANSITION/LATENT over $\hat{S}(z)$	Input dist. × task structure × annotation protocol × deployment context
Proxy	$\hat{S}(z)$: blind to local environments	Accuracy/BLEU: blind to calibration, tail behaviour, cross-regime transfer
Placement	Physical (z, k), instrument (BAO/$f\sigma_8$/CMB), interpretive	Input domain, evaluation instrument, interpretive framework
Episenter	z-space vs. S-space: different residual structures	Accuracy vs. calibration error: different model rankings
Compression	$P(k)$ discards phase and higher-order correlations	Scalar score discards variance, tail structure, failure modes

Table 7: Structural parallel across both domains. All five framework concepts map onto both domains non-trivially. The parallel is not metaphorical—it reflects the same measurement-theoretic structure operating in both cases.

5.7 Validity-Aware Evaluation in Practice

A validity-aware evaluation requires five declarations:

1. **Regime declaration:** Every result must be labelled with its evaluation regime (input distribution, task structure, annotation protocol, deployment context). Results are regime-scoped, not universal.
2. **Proxy declaration:** Every metric must be accompanied by its target quantity, visibility scope, and structural blindness.
3. **Placement specification:** Instrument, input, and interpretive placements must be declared.
4. **Episenter transparency:** The primary metric must be declared as a choice. Alternative episenter placements must be assessed and their divergence from the primary episenter reported.
5. **Compression accounting:** The report must identify what the chosen compression retains and what it discards, and how consequential the loss is for the intended deployment.

Validity profile (minimal structure):
Model: [identifier]
Evaluation regime: [input distribution + task structure + annotation + deployment]
Primary metric: [name] — proxy for [target] — declared blindness: [list]

Regime-validity: valid within specified regime; transfer validity: [tested / unknown]
Episenter: [primary metric] — alternatives assessed: [list with divergence]
Compression loss: following dimensions not captured: [list] — consequentiality: [assessment]

5.8 Honest Scope Statement for Case 2

This case does not claim that current AI evaluation is fundamentally broken or that existing benchmarks are useless. It claims that current evaluation is regime-implicit, proxy-opaque, episenter-undeclared, and compression-unaccounted— and that these failures produce predictable errors in specific conditions. The validity-aware framework does not replace existing evaluation; it provides a diagnostic language for understanding when and why existing evaluation fails, and a minimal structure for making evaluation results more epistemically honest.

5.9 Testable Predictions from Case 2

A1 Models evaluated near benchmark regime boundaries show higher output variance and lower metric reliability than models at regime centres. *Status: Methodological—testable on any existing benchmark + OOD dataset.*

A2 Substituting accuracy with calibration error produces systematic, directional rank changes: models with high accuracy and poor calibration fall; models with lower accuracy and better calibration rise. *Status: Methodological—testable on MMLU + calibration data.*

A3 Cross-regime Spearman $r < 0.5$ (formal-academic vs. conversational/adversarial); within-regime $r > 0.75$. *Status: ACTIVE—data available.*

A4 SHAP coherence in training regime is uncorrelated with OOD performance. *Status: Methodological.*

A5 Validity profile (regime-scoped) reduces deployment failure prediction error by $> 20\%$ vs. scalar score alone. *Status: PENDING—requires production deployment incident dataset.*

5.10 What This Framework Adds Beyond Standard Evaluation Theory

Standard evaluation theory in machine learning identifies many of the failure modes described above—distributional shift, benchmark overfitting, metric gaming. What the present framework adds is not a list of warnings, but a *formal structure* for locating them.

The key contribution is the claim that evaluation regime, proxy, placement, episenter, and compression are not independent design choices to be managed case by case—they form a structured chain (regime → proxy+placement → episenter → compression → output) with predictable, cross-domain failure modes. This is not a novel observation about AI evaluation specifically; it is an application of a general measurement-theoretic architecture that holds equally in cosmological structure analysis.

The claim that both domains instantiate the *same formal object*—not an analogy but a shared measurement-theoretic structure—is the specific contribution that distinguishes this framework from domain-level evaluation caution. A validity profile is not a checklist; it is a coordinate in the framework's parameter space, with the same structure as a cartographic coordinate in $R(k, S)$. The prediction register for AI (A1–A5) follows the same derivation logic as the cosmological prediction register (C1–C6). That is not metaphor; it is the content of the framework.

6 Synthesis: Cartography as Methodological Imperative

The two cases share one structural finding: treating observed output as a direct read of the underlying phenomenon produces predictable, diagnosable errors with the specific signature of the mediating factors that were ignored.

6.1 What the Cases Establish Together

The structural parallel between Case 1 and Case 2 is the central argument for the framework's generality. The five framework concepts map onto both domains non-trivially, and the failure modes they identify are specific and named. This is not a coincidence of domain selection—it reflects the claim that the framework identifies general features of mediated measurement in complex systems.

A failure mode in one domain has a structural analogue in the other:

- Proxy conflation in cosmology (treating z as a direct read of structural state) has a direct analogue in AI (treating BLEU as a direct read of translation quality).
- Episenter universalism in cosmology (placing H_0 at the centre of all inference) has a direct analogue in AI (placing accuracy at the centre of all evaluation).
- The cartographic reframing in cosmology ($R(k, S)$ response surface) has a direct analogue in AI (validity profile as a regime-scoped coordinate map).

6.2 Cartography Is Not Relativism

The cartographic reframing does not claim that all measurements are equally valid or that truth is inaccessible. It claims that measurements are coordinates in a structured space, and that the structure of that space must be made explicit before measurements can be meaningfully compared.

A coordinate is not less real for being regime-conditioned—it is more precisely located. A measurement with declared proxy, placement, episenter, and compression accounting is *more* informative than a measurement without these declarations, not less.

6.3 Regime Identification as Falsifiable Procedure

Regime identification is not circular. Declaring that two measurements come from different regimes is a *hypothesis*, not an escape hatch. The hypothesis predicts that making the measurements regime-compatible will change the apparent conflict.

If the conflict persists after full placement compatibility is established—if regime-compatible CMB and SNIa measurements still show the same H_0 tension—then the regime-mismatch explanation is falsified. This is not a theoretical escape from falsification; it is a specific, empirically testable prediction.

6.4 The Higher Standard

The cartographic framework demands more from analysis, not less. A theory-agnostic formalisation of the Structural Coherence Evaluation approach underlying this demand is developed in [11]. The requirements are:

- Regime declaration before any cross-regime comparison.
- Proxy construction with explicitly declared limitations.
- Placement specification across all three dimensions.
- Episenter transparency and alternative-episenter assessment.
- Compression accounting in every evaluation report.

Meeting this standard is more expensive than the standard single-measurement, single-model, single-episenter analysis. The payoff is regime-stable results that can be compared across studies, reproduced across instruments, and falsified with precision.

7 Implications: Intelligence, Consciousness, Observer Proximity

Scope statement: This section presents implications of the framework, not premises. The claims here are downstream of the framework's empirical content, not foundational to it. They are offered as directions for future work, not as established results. The framework's scientific validity does not depend on any claim made in this section.

7.1 Intelligence as a Compression System

If compression is a general feature of measurement, then any system that processes information—including biological and artificial intelligence—is a compression system operating within a regime. The question "how intelligent is this system?" cannot be answered without specifying: (a) the evaluation regime, (b) the proxy used to make intelligence measurable, (c) the observer's placement, and (d) what the compression retains and discards.

This is not a metaphor. The validity-aware AI framework in Section 5 is a direct application of this implication. The framework's predictions about AI evaluation (A1–A5) are regime-conditioned, proxy-explicit, and episenter-declared. They follow directly from treating intelligence measurement as a special case of regime-bound measurement.

The implication extends to biological intelligence. Measurements of cognitive capability—IQ tests, neuropsychological batteries, learning rate assessments—are all regime-conditioned proxies with declared and undeclared structural blindness. The validity-aware framework applies to them with exactly the same structure as it applies to AI benchmarks.

7.2 Observer Proximity and Distance

The framework provides a structural language for what it means for an observer to be near or far from a phenomenon. Proximity is not just physical distance—it is a function of proxy choice, regime compatibility, and placement alignment.

An observer is *near* to a phenomenon (in the measurement-theoretic sense) when:

- Their proxy captures the phenomenon's most relevant structural dimensions;
- They are operating within the phenomenon's regime;
- Their placement is compatible with the phenomenon's characteristic scales and structures;
- Their episenter organises the analysis around the phenomenon's primary dimensions of variation.

An observer is *far* when they use a proxy blind to the relevant dimensions, or whose episenter organises the analysis around the wrong variables—regardless of physical proximity. A physicist at $z = 1100$ observing the CMB is far from the LATENT-regime dark energy dynamics in the sense that their measurement placement is not sensitive to LATENT-regime structure. A galaxy survey at $z = 0.5$–0.8 is near to TRANSITION-regime dynamics.

This structural notion of proximity/distance provides a language for interpreting disagreements between observers that goes beyond the usual dichotomy of "right vs. wrong". Many disagreements that appear to be about empirical content are actually about proxy choice, episenter placement, or regime scope. Making these factors explicit is a precondition for productive disagreement.

7.3 Consciousness as a Boundary Case

The framework implies a structural question about consciousness, without making a claim about it. The question is: is consciousness a measurement instrument, a regime boundary condition, or a phenomenon to be measured?

If it is an instrument—if what is observed depends on the observer's conscious state—then the framework requires that conscious observation be treated as a placement variable, not a neutral background condition. This is not a claim about consciousness. It is an implication of the measurement framework that points toward a research question: under what conditions does the observer's conscious state constitute a placement variable that affects what is seen?

8 Testable Predictions and Falsification Criteria

A framework that cannot be falsified is not a scientific framework. The purpose of this section is to make the framework falsifiable: to state what it predicts, what would break it, how its predictions differ from competing explanations, and what the current evidence status is.

8.1 Four Criteria for a Genuine Test

C1 — Pre-specification: The prediction must be stated before the test is run, with the test protocol locked. Post-hoc reframing of an observation as "consistent with the framework" is not a test.

C2 — Discriminability: The prediction must be distinguishable from the predictions of at least one competing explanation. A prediction consistent with all plausible alternatives contributes no evidential weight.

C3 — Directionality: The prediction must specify a direction—not just "an effect will be observed" but "the effect will be in this direction, of approximately this magnitude, under these conditions".

C4 — Regime-scope declaration: The prediction must declare the regime in which it holds. A prediction untestable because the regime was never specified is not a scientific prediction.

8.2 Complete Prediction Register

8.3 Differentiation from Competing Explanations

Cosmology: regime structure vs. ΛCDM + systematics

AI evaluation: validity-aware vs. standard benchmark practice

8.4 Framework-Breaking Outcomes

F-BREAK 1: Regime-random residuals. LATENT, TRANSITION, and FLOW regime residuals are statistically indistinguishable. Sign coherence across regimes is not significantly different from random. The regime structure hypothesis is falsified. The framework requires that regimes produce systematically different residual patterns.

F-BREAK 2: Proxy-substitution drift is random. The direction of drift after proxy substitution cannot be predicted from the proxy's declared structural blindness—the drift is indistinguishable from a random permutation. The proxy theory is falsified.

F-BREAK 3: Episenter shift produces random redistribution. Rank changes after metric substitution in AI evaluation are not predictable from models' known accuracy-vs-calibration profiles—the redistribution is indistinguishable from random reordering. The episenter theory is falsified.

F-BREAK 4: Hubble tension survives full regime-compatibility test. The tension persists at full statistical significance after all three placement dimensions (physical, instrument, interpretive) have been made regime-compatible. The regime-

ID	Falsifiable prediction	Test protocol	Dom.	Status
COSMOLOGY — EFC/WP3 Regime-Bound Predictions				
C1	Same (k, S) coordinate yields consistent response across independent probes (BAO, $f\sigma_8$, lensing)	Map DESI DR2 + $f\sigma_8$ onto (k, S); test consistency vs. probe-specific null	Cosm.	PARTIAL
C2	Proxy substitution from global $\hat{S}$ to local structural proxy yields directional drift; direction predictable from declared blindness	Compare regime assignments: $\hat{S}(z)$ vs. morphology proxy on SDSS/GAMA	Cosm.	Pending
C3	Largest BAO/growth deviations cluster near FLOW/TRANSITION and TRANSITION/LATENT boundaries in $\hat{S}$-space. DESI $z = 1.0$ boundary coincidence holds at full survey statistics	Axiom 0 v2 pre-registered on DESI Year-5 ($N \geq 30$); sign coherence as primary stat	Cosm.	ACTIVE
C4	H_0 within a single $\hat{S}$-regime shows lower inter-survey variance than pooled cross-regime estimates	Stratify existing H_0 measurements by $\hat{S}$-regime; compute within- vs. cross-regime variance	Cosm.	ACTIVE
C5	LATENT residuals sign-coherent and low-variance; FLOW residuals higher-variance; ratio > 2.0	Compute residuals per regime; runs-test + sign-coherence + variance ratio vs. permutation null	Cosm.	PARTIAL
C6	Models matched on χ^2_{BAO} but differing in proxy construction diverge in $f\sigma_8$ predictions at TRANSITION coordinates ($z =$ 0.5–0.8)	Select matched model pairs; evaluate $f\sigma_8$ at $\hat{S}$-identified TRANSITION coords	Cosm.	Pending
AI EVALUATION — Validity-Aware Framework Predictions				
A1	Models near benchmark regime boundaries show higher output variance (boundary distance predicts variance increase, $r >$ 0.4)	Sample inputs at graduated OOD distance; compute output variance per distance band; test monotonicity	AI	Meth.
A2	Accuracy → calibration error substitution: top-quartile-accuracy/bottom-half-calibration models fall ≥ 2 quartiles	Apply both metrics to MMLU; compute rank changes; test direction vs. permutation null	AI	Meth.
A3	Spearman $r < 0.5$ between formal-academic and conversational/adversarial rankings; within-regime $r > 0.75$	MMLU vs. MT-Bench / adversarial NLI ranking correlation	AI	ACTIVE
A4	SHAP coherence in training regime is uncorrelated with OOD performance (calibration and accuracy)	Compute SHAP coherence on in-dist.; evaluate same models OOD; test correlation	AI	Meth.
A5	Validity profile reduces deployment failure prediction error by $> 20\%$ vs. scalar score alone	Collect production incident data; compare prediction accuracy (a) vs. (b)	AI	Pending
FRAMEWORK-LEVEL — Cross-Domain Structural Predictions				
F1	Proxy substitution drift direction in both domains is predictable from declared structural blindness (not random)	Proxy swap experiments in both domains; test direction vs. permutation null	Both	ACTIVE
F2	Conflicts resolved by regime identification but not by model substitution are more frequent than the reverse	Systematic conflict taxonomy in both domains; resolution scoring	Both	Meth.

Table 8: Complete prediction register (13 predictions). Criteria C1–C4 from Section 8.1 apply to all predictions. Status: ACTIVE = pre-registered, awaiting data; PARTIAL = preliminary result, not confirmatory; Pending = dataset not yet available; Meth. = methodological, testable on existing data, analysis not yet run.

Prediction	**This framework**	**ΛCDM + systematics**
Residual structure by regime	Residuals sign-coherent within TRANSITION/LATENT; variance increases near boundaries	Residuals uniform across z; boundary clustering is coincidental
Hubble tension resolution	Tension reduces when probes are made regime-compatible	Tension persists; requires new physics
Proxy substitution	Systematic, directional drift; direction predictable from proxy blindness	Proxy substitution produces noise; no systematic direction
Null result distribution	Null results concentrated in LATENT regime	Null results distributed randomly across z
H_0 within-regime variance	H_0 within one regime has lower variance than pooled	Inter-survey variance is instrument-dependent, not regime-dependent

Table 9: Differentiation: framework predictions vs. ΛCDM + systematics. Each row is a discriminating prediction.

Prediction	**This framework**	**Standard benchmark practice**
Cross-regime rank correlation	Spearman $r < 0.5$ between formal-academic and adversarial/conversational regimes	Rankings reflect general capability; $r > 0.75$ for strong models
Episenter shift effect	Systematic, directional rank changes when accuracy → calibration error	Rankings robust to metric choice; changes are noise
Explainability–validity link	SHAP coherence in training regime uncorrelated with OOD performance	High explainability predicts better transfer performance
Validity profile utility	Regime-scoped evaluation reduces deployment failure prediction error $> 20\%$	Benchmark score sufficient predictor of deployment performance

Table 10: Differentiation: validity-aware framework vs. standard benchmark practice.

mismatch explanation is falsified; the tension must be attributed to physics beyond the framework's explanatory scope.

F-BREAK 5: Null results are regime-random. Null results are distributed uniformly across regimes. LATENT does not show higher null-result concentration than FLOW or TRANSITION. The cartographic null-result prediction is falsified.

8.5 Partial-Revision Triggers

The following outcomes would not break the framework but would require targeted revision of specific components:

- $\hat{S}(z)$ fails to predict regime boundaries at $z > 1.5$: requires extending or replacing the proxy construction for high-z, not abandoning regime structure.
- Sign coherence does not replicate at $N > 30$: requires revising the test design; regime hypothesis could still hold with a different test statistic.
- Cross-regime rank correlation in AI is domain-pair-dependent (some pairs show $r > 0.75$): requires refining the regime boundary definition, not abandoning the regime concept.
- Episenter shift produces predicted redistribution at smaller magnitude than expected: requires recalibrating quantitative predictions; structural prediction confirmed.

8.6 What the Framework Cannot Predict

- **The value of H_0 in any regime:** The framework predicts that H_0 is regime-conditioned. What the regime-specific values are is a question for the underlying physics, not for the measurement framework.
- **Which AI model is best:** The framework predicts that "best" is regime-conditioned. Which model is best in which regime is a question for the evaluation data.
- **Resolution of the Hubble tension:** The framework provides a diagnostic reframing. Whether the tension survives a full regime-compatibility analysis is an empirical question.
- **The number of regimes in any domain:** The framework requires that regimes be defined before the test. The number and location of regime boundaries is determined by the domain physics and proxy construction.

8.7 Falsification Protocol

For a result to count as a falsification (not merely a null result), three conditions must be met:

> **FC1 — Pre-registered test:** The test must have been pre-registered with the prediction and protocol specified before execution.
>
> **FC2 — Adequate power:** The test must have adequate statistical power to detect the predicted effect at the stated magnitude.
>
> **FC3 — Regime-compatible execution:** The test must be executed in the regime the prediction specifies. A null in the wrong regime does not falsify a regime-scoped prediction.

A result meeting all three conditions and producing a statistically significant outcome in the *opposite* direction to the prediction constitutes a falsification.

8.8 Current Evidence Status Summary

ID	Prediction	Status	Current evidence
C1	Probe consistency at (k, S)	PARTIAL	$p = 0.020$ sign coherence, $N = 10$, secondary statistic
C2	Proxy substitution drift direction	Pending	Test designed; local proxy not yet available
C3	Boundary clustering (primary)	ACTIVE	DESI $z = 1.0$ dist$= 0.000$ (anecdotal at $N = 10$)
C4	H_0 within-regime variance	ACTIVE	Analysis not yet run
C5	LATENT residual sign coherence	PARTIAL	8/9 positive sign coherence in TRANSITION+LATENT
C6	Proxy divergence at TRANSITION	Pending	Requires Euclid WL + DESI growth sample
A1	Boundary variance increase	Meth.	Testable on any benchmark + OOD dataset
A2	Episenter shift redistribution	Meth.	Testable on MMLU + calibration data
A3	Cross-regime rank correlation	ACTIVE	Data available; analysis not yet published
A4	Explainability–validity decoupling	Meth.	Requires paired in-dist./OOD evaluation
A5	Validity profile utility	Pending	Requires production incident dataset
F1	Proxy drift directionality (both)	ACTIVE	Cosmology component partially run (Axiom 0)
F2	Regime-ID resolution frequency	Meth.	Requires systematic conflict taxonomy

Table 11: Current evidence status for all 13 predictions. Legend: ACTIVE = pre-registered, awaiting data; PARTIAL = preliminary result, not confirmatory; Pending = dataset not yet available; Meth. = analysis not yet run.

9 Conclusion

The measurement framework developed in this paper rests on one claim: observed outputs in complex systems are not direct reads of the underlying phenomenon, but regime-conditioned products of proxy choice, instrument logic, observer placement, and interpretive compression.

9.1 What the Framework Establishes

The framework establishes four things:

1. **A conceptual architecture:** Five operative concepts (regime, proxy, placement, episenter, compression) that are jointly sufficient to characterise the epistemic structure of any measurement system in a complex domain.

2. **A diagnostic language:** Three named failure modes (regime conflation, proxy opacity, episenter universalism) with specific signatures and specific remedies.

3. **A falsifiable prediction structure:** Thirteen predictions across two domains, each satisfying the four genuineness criteria (pre-specification, discriminability, directionality, regime-scope). Five of these are currently ACTIVE; five are methodological (testable on existing data); three are pending specific datasets.

4. **Cross-domain generality:** All five framework concepts map onto both cosmological structure analysis and AI evaluation in a non-trivial way. This is the central argument that the framework identifies general features of mediated measurement, not features of

a specific domain.

9.2 Contributions of the Two Cases

The cosmological case (WP3/$R(k, S)$) demonstrated that the Hubble tension and related BAO anomalies can be reframed as placement and regime questions before they are treated as signals of new physics. The structural proxy $\hat{S}(z)$ was constructed with declared limitations and a locked parametrisation. Axiom 0 produced a preliminary sign-coherence result ($p = 0.020$, $N = 10$, secondary statistic) consistent with EFC regime structure but not yet confirmatory. The DESI $z = 1.0$ boundary coincidence (dist$= 0.000$) is a cartographic observation requiring larger sample size to interpret.

The AI evaluation case demonstrated that standard benchmark practice exhibits the three named failures. The structural parallel between the two domains—a five-concept table that maps non-trivially onto both—is the central argument for the framework's generality. The validity-aware evaluation framework provides a direct, operational remedy for each failure.

9.3 What Remains

The four highest-priority open items:

1. **Axiom 0 v2:** Pre-registered primary test on DESI Year-5 data, with sign-coherence as the primary statistic and $N \geq 30$.
2. **Regime-stratified H_0 analysis:** Apply the placement-compatibility framework to existing H_0 literature; test C4.
3. **Cross-regime AI rank correlation:** Test A3 on MMLU vs. MT-Bench/adversarial NLI with existing model data.
4. **Proxy substitution test (both domains):** Test F1—verify that proxy substitution drift direction is predictable from declared structural blindness.

These are analyses that can be run on existing or near-term data. The framework has a falsification surface. The predictions are stated. The question is whether they hold.

A parallel development on the field-theoretic side—deriving the relativistic perturbation-sector implications of EFC (the μ–Σ split required by the survival valley) from a minimal effective field theory ansatz—is documented in [14]. That paper and this one are structurally complementary: this paper specifies the measurement conditions for regime-stable claims; that paper specifies the field-level derivations that must produce the regime structure.

Competing interests: None declared.
Data availability: WP3 data and Axiom 0 test scripts available on request.
Companion documents: Full Section 2, Section 4, Section 5, and Section 8 working documents available separately. The complete EFC Validation Ledger (v3.3) is publicly accessible at `https://supertedai.github.io/EFC/public/EFC_Validation_Ledger.html`.
Version: Working paper — March 7, 2026

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